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Sex- and Trimester-Specific Associations of Prenatal Co-Exposure to Organophosphate Esters and Phthalates with Preschoolers' Trajectories of Co-Occurring ADHD and ASD Symptoms: Cord Blood Metabolomic Study in the Ma'anshan Birth Cohort

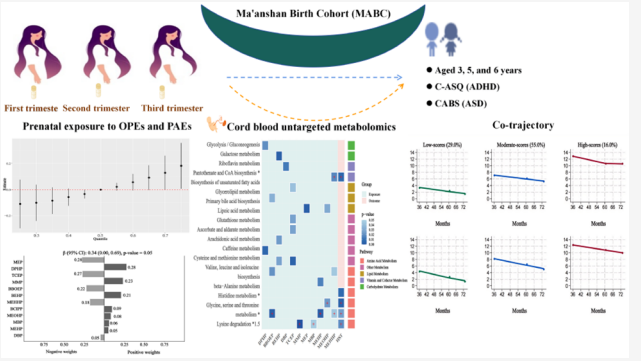
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ABSTRACT: Although neurotoxic, prenatal exposure to organophosphate esters (OPEs) and phthalic acid esters (PAEs) and their effects on preschoolers' autism spectrum disorder (ASD) and attention-deficit hyperactivity disorder (ADHD) cotrajectories and underlying metabolic mechanisms remain unclear, we aimed to elucidate these links. Maternal urinary OPEs/PAEs were measured in 3040 dyads from the Ma'anshan Birth Cohort across three trimesters. Child ADHD/ASD symptom scale scores were assessed at ages 3, 5, and 6, and cotrajectories were identified using group-based multitrajectory modeling. Single-pollutant models revealed that bis(2-ethylhexyl) phosphate (BEHP) across pregnancy was positively associated with high-score trajectories (HST) (OR = 1.20, 95% CI: 1.06, 1.37), whereas bis(2-butoxyethyl) phosphate (BBOEP) exhibited U-shaped associations. Second-trimester diphenyl phosphate (DPHP) (OR = 1.13, 95% CI: 1.01, 1.26), BEHP (OR = 1.14, 95% CI: 1.04, 1.24), and monobutyl phthalate (OR = 1.15, 95% CI: 1.00, 1.32) were positively associated with HST. First-trimester DPHP exhibited a positive correlation with moderate-score trajectories and HST in girls, while bis(1-chloro-2-propyl) phosphate across pregnancy was inversely associated with HST in boys ($p_{sex\text{-}int} < 0.05$). No mixed effects were detected. BBOEP across pregnancy was negatively associated with ADHD symptoms, whereas BEHP was positively associated. BEHP, monomethyl phthalate, and mono-(2-ethyl-5-oxohexyl) phthalate were positively associated with ASD symptoms, whereas dibutyl phosphate and monoethyl phthalate were negatively associated ($p < 0.05$). Cord blood metabolomics identified pyrimidine, biotin, lysine, cysteine, and methionine metabolism as key mediators of OPE-induced cotrajectories, and purine metabolism mediated PAEs' effects ($p < 0.05$). This study highlights OPE/PAE neurotoxicity and reveals novel cord metabolomic insights.



KEYWORDS: *endocrine-disrupting chemicals, autism spectrum disorder, attention-deficit hyperactivity disorder, trajectory, metabolomics*

1. INTRODUCTION

Attention-deficit hyperactivity disorder (ADHD) and autism spectrum disorder (ASD) are common neurodevelopmental disorders in children, and their prevalence is increasing;[1−3] symptoms and functional impairments often persist into adulthood.[4] Despite differences in clinical manifestations, overlapping symptoms and heritability have been reported, suggesting shared developmental pathways.[5] In one study, 13% adolescents with ADHD were diagnosed with comorbid ASD.[6] Other studies have reported that approximately one in eight adolescents with ADHD has ASD.[7] ADHD is also among the most common comorbid conditions in patients with ASD, with a comorbidity rate ranging from 40% to 70%.[8] Co-occurring disorders not only complicate diagnosis and rehabilitation but also affect prognosis.[9] The high co-occurrence rate between ADHD and ASD stems from potential shared pathophysiological mechanisms, as both can influence brain neurodevelopment through similar genetic, immune, gut microbiota, and environmental pathways.[10] Notably, individuals who present with partial disease symptoms or subclinical functional deficits (e.g., ADHD and ASD symptoms) are far more

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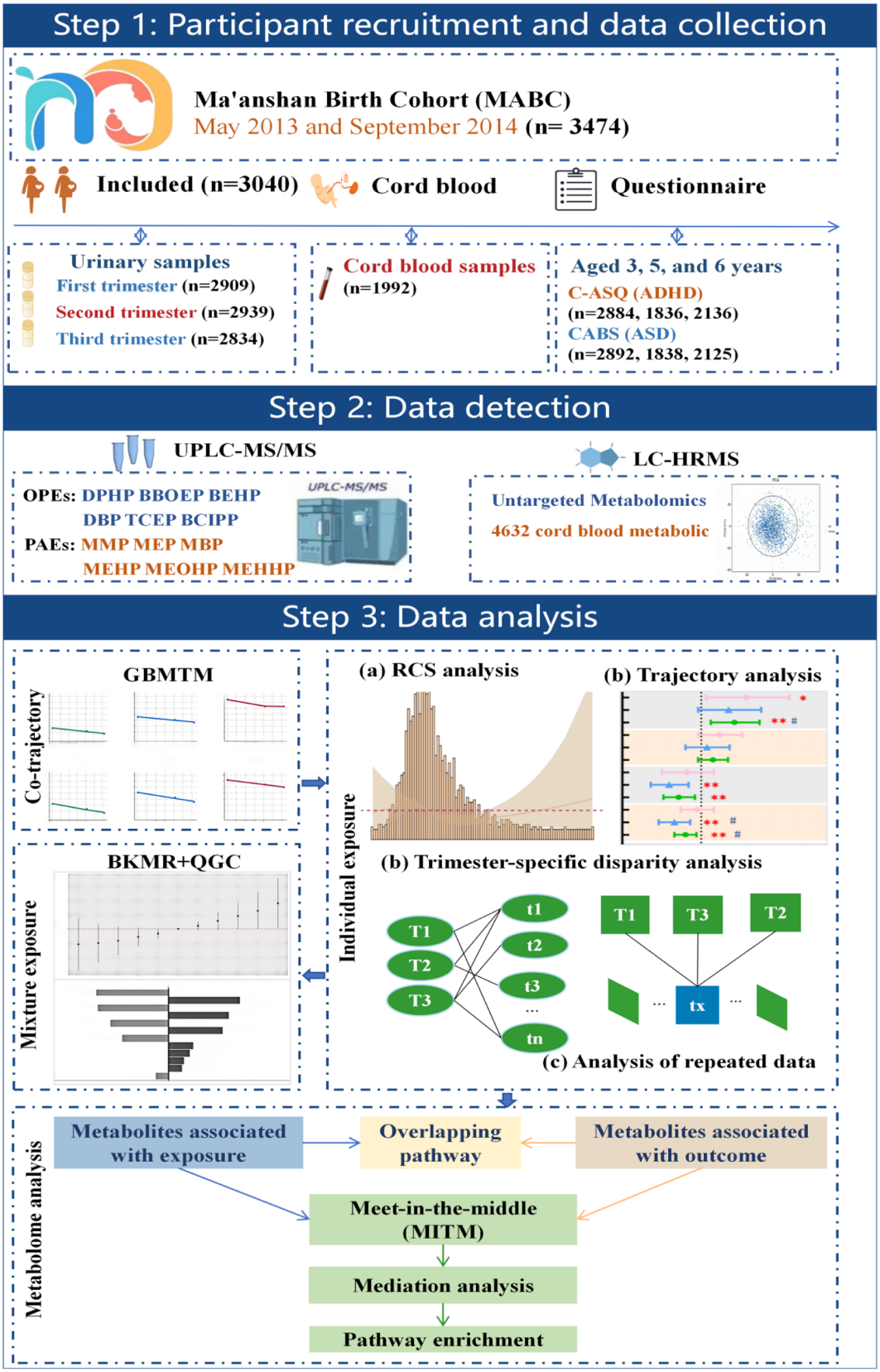


Figure 1. Flowchart showing the overall study design.

common than those with clinically diagnosed mental disorders.[11] This observation suggests that early identification and treatment of ADHD and ASD during childhood can significantly improve long-term health outcomes.

The etiology of ADHD and ASD remains unclear, and organophosphate esters (OPEs) and phthalic acid esters (PAEs) are two standard classes of emerging environmental chemicals with increasingly recognized neurotoxicity. OPEs are

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widely used, halogen-free flame retardants in industrial and consumer goods (e.g., furniture, electronics).[12,13] PAEs are synthetic endocrine-disrupting plasticizers added to polymers (e.g., in food packaging and medical devices) to enhance flexibility and durability.[14] Owing to weak interactions between OPEs and PAEs, and between these materials and other materials during production (van der Waals forces), they readily migrate from polymers into the environment.[14,15] The prenatal period is a phase of rapid fetal development, during which the fetus is susceptible to damage from exogenous stimuli. Current research has detected OPEs in placental tissues at the maternal−fetal interface, including the chorion and decidua.[16,17] Similarly, PAEs have been identified in amniotic fluid and cord serum.[18] OPEs and PAEs share numerous similarities in terms of their exposure pathways and health effects.[14] When humans are simultaneously exposed to both substances, the combined effects may be more complex and pronounced.[19] Therefore, combined exposure to OPEs and PAEs warrants serious attention, particularly during critical periods of fetal neurodevelopment in utero.

Cunha et al. systematically reviewed 8 cohort studies on prenatal PAE and ASD: 4 reported a positive association, and 4 reported no significant association.[20] A Chinese study indicated that prenatal exposure to metals and PAEs may increase the risk of ASD at age four by 2.11-fold.[21] A cohort study reported that diphenyl phosphate (DPHP) and bis(1,3-dichloro-2-propyl) phosphate (BDCIPP) in the second trimester were associated with increased ADHD risk in children, with girls more susceptible to DPHP.[22] Hall et al. reported that prenatal exposure to di-*n*-butyl phosphate and BDCIPP may increase the risk of ADHD in offspring. And bis(2-butoxyethyl) phosphate (BBOEP) exhibited a sex-specific, risk increased with high exposure in boys, whereas it decreased in girls.[23] A U.S. birth cohort study demonstrated significant positive correlations between maternal DPHP and BDCIPP and hyperactivity-impulsivity, attention problems, and externalizing behavior problems in 3-year-old children, and that girls' cognitive and behavioral development is more susceptible to BDCIPP exposure.[24] These studies suggest that prenatal exposure to OPEs/PAEs may affect offspring neurodevelopment, with potential sex-and trimester-specific effects.

Nevertheless, few studies have examined the role of cord blood metabolomics in the association between prenatal chemical exposure and offspring neurodevelopmental disorders.[25−27] Cord blood metabolomics closely mirrors fetal metabolic status and is widely used to characterize biological perturbations.[27] Previous studies have linked alterations in cord blood metabolomics to ADHD/ASD. For instance, linolenic acid in cord serum is negatively associated with ASD, whereas perfluorooctanoic acid shows the strongest positive association.[28] A prospective birth cohort study demonstrated that elevated levels of branched-chain amino acids in cord plasma are associated with an increased risk of ADHD.[29] Additionally, research has revealed that cord blood plasma-chained 5-methoxytryptophol is negatively correlated with the risk of both ASD and ADHD, and that tryptophan and other metabolites were associated with subsequent ADHD risk.[30] Prenatal OPEs and PAEs have also been shown to alter the profile of neonatal metabolites, particularly in the tyrosine and tryptophan pathways, which are associated with infant neurobehavioral outcomes.[15,25,27] By identifying metabolites related to environmental exposure and disease, metabolomics-wide association studies (MWASs)—especially when combined with the "meet-in-the-middle" (MITM) approach—offer a promising strategy for uncovering key molecular mechanisms linked to health effects.[27]

Based on the Ma'anshan Birth Cohort (MABC), with multiple time-point repeated measurements of maternal urinary exposure and childhood assessments of ADHD and ASD symptoms. This study aims to (1) investigate the relationship between individual and combined exposure to OPEs and PAEs and the cotrajectory of ADHD-ASD symptoms; (2) identify sensitive exposure windows and sex-specific disparities in prenatal exposure to OPEs and PAEs; and (3) explore potential cord blood metabolic pathways linking coexposure to OPEs and PAEs to the cotrajectory by integrating the MWAS and MITM methods. The overall study design is shown in Figure 1.

2. METHODS

2.1. Study Population

We used data from the MABC,[31] involving 3226 pregnant women who provided at least one urine sample during the three trimesters. We further excluded 186 individuals whose ASD and ADHD symptom questionnaires were completed fewer than once. Ultimately, our study included 3040 mother−child dyads, with cord blood samples available for 1992 individuals, who underwent metabolomics analysis (Figure S1). The number of individuals in the first, second, and third trimesters was 2909, 2939, and 2834, respectively. The study was approved by the Ethics and Research Committees of Anhui Medical University (No. 20131195), and all participants provided written informed consent before enrollment and follow-up.

2.2. Measurement of Urine OPEs and PAEs

Maternal morning urine samples collected during each trimester were aliquoted into 50 mL polypropylene urine cups, transferred to 5 mL polypropylene cryovials, and stored at − 80 °C until measurement.

On the basis of the initial exploration of OPE detection methods and a literature review, we selected an OPE [tris(2-chloroethyl) phosphate (TCEP)] and five OPE metabolites—DPHP, bis(1-chloro-2-propyl) phosphate (BCIPP), BBOEP, bis(2-ethylhexyl) phosphate (BEHP), and dibutyl phosphate (DBP)—for urine testing during pregnancy. These metabolites are relatively common and have relatively high detection rates.[32] The six OPEs were detected in urine samples using solvent-induced phase change extraction (SIPTE) and ultraperformance liquid chromatography−tandem mass spectrometry (UPLC−MS/MS) (Xevo TQ−SLC−MS, Waters Corporation, 34 Maple Street, Milford, MA, USA) with an isotope internal standard assay.[32] UPLC was performed on an ACQUITY C18 BEH column (50 mm × 2.1 mm × 1.7 μm) from Waters Corporation (Milford, MA, USA). Several quality assurance and control (QA/QC) measures were followed to ensure accurate quantification. Additional details of the urine OPE measurements and QA/QC procedures are available in Text S1 and Tables S1−S2. The detailed experimental method for the OPEs has been previously described.[13,32] The limits of detection (LODs) for OPEs are listed in Table S2.

Similarly, six target PAEs—monomethyl phthalate (MMP), monoethyl phthalate (MEP), monobutyl phthalate (MBP), mono-2-ethylhexyl phthalate (MEHP), mono-(2-ethyl-5-hydroxyhexyl) phthalate (MEHHP), and mono-(2-ethyl-5-oxohexyl) phthalate (MEOHP)—were measured using previously published methods.[33] The concentrations of PAEs in the urine samples were quantified using an isotopic internal standard method, namely, solid-phase extraction and high-performance liquid chromatography−tandem mass spectrometry (6410 LC−MS, Agilent Technologies, Santa Clara, CA, USA).[14,15,33] QA/QC measures were implemented to ensure accurate quantification of PAEs. Detailed information on urine PAE measurements and QA/QC procedures is provided in Text S2 and Tables S3−S4. The LODs for the PAEs are listed in Table S4.

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To account for individual urine dilution, a commercial assay kit (Jiancheng Bioengineering Institute, Nanjing, China) was used to measure urinary creatinine (CR) concentrations via the picric acid method, followed by creatinine correction for OPE and PAE concentrations in urine, according to the formula detailed in our previous study.[32]

2.3. Cord Blood Metabolome Profiling

Untargeted metabolomic profiling of cord blood serum was performed using a liquid chromatography−high-resolution mass spectrometry (LC−HRMS) system comprising a Waters ACQUITY UPLC I-Class Plus ultraperformance liquid chromatography system and a Thermo QE HF high-resolution mass spectrometer, operating in both positive and negative ion modes. Detailed methods, including sample extraction, quality control procedures, UPLC MS/MS settings, and data annotation workflows, are provided in Text S3.

2.4. Assessment of ADHD and ASD Symptoms

The Chinese version of the Conners Abbreviated Symptom Questionnaire (C-ASQ) was used to assess ADHD symptoms in children aged 3, 5, and 6 years. The C-ASQ included 10 items, with responses ranging from "none" (1) to "a lot" (4). Parents completed the C-ASQ for each age group; a total score ≥15 indicated ADHD symptoms. The C-ASQ is adapted from the Revised Conners Parent Rating Scale.[34,35] As previously reported, its sensitivity, accuracy, and specificity are 76.0%, 82.1%, and 92.9%, respectively.[34] The internal consistency coefficients (Cronbach's α) at ages 3, 5, and 6 were 0.871, 0.845, and 0.878, respectively. The number of ADHD symptom questionnaires at ages 3, 5, and 6 was 2884, 1836, and 2136, respectively.

The Clancy Autism Behavior Scale (C-ABS) is the most widely used tool for assessing ASD symptoms. The C-ABS includes 14 items completed by parents or guardians. In 1983, researchers in China revised it and adopted a three-point scoring system, replacing the options "yes" and "no" with "never," "occasionally," and "often," assigning scores of 0, 1, and 2, respectively. A total score ≥14 indicated suspected ASD symptoms. In Chinese children, the sensitivity, specificity, and positive predictive value are 58%, 84%, and 65%.[33] Cronbach's α was 0.802, 0.798, and 0.853 at ages 3, 5, and 6 years, respectively. The number for the ASD symptom questionnaire at ages 3, 5, and 6 was 2892, 1838, and 2125, respectively.

2.5. Covariates and Other Information

In our study, trained on-site staff collected basic information using the self-designed Maternal and Infant Health Record Form during women's prenatal checkups. Moreover, pregnancy complications, mode of delivery, fetal birth outcomes, and fetal sex were obtained from the hospital's electronic medical record system. The intelligence quotient (IQ) of pregnant women was assessed by trained research staff using the Wechsler Adult Intelligence Scale-IV subtest.[13]

On the basis of previous studies[13,25,34] and the use of a directed acyclic graph (DAG) to identify potential covariates, the final covariates were as follows: maternal age, prepregnancy body mass index (BMI), gestational weight gain, per capita monthly family income, maternal educational level, delivery mode, parity, maternal IQ, infant sex, and feeding mode, as detailed in Figure S2.

2.6. Statistical Analysis

We present quantitative data as the mean ± standard deviation and qualitative data as frequency and composition ratios. The concentrations of OPEs and PAEs below the LOD were estimated as LOD/$\sqrt{2}$.[36] Intragroup correlation coefficients (ICCs) for OPEs and PAEs before and after CR-adjustment were used to assess reproducibility, which revealed that all OPEs and PAEs exhibited high temporal variability (ICCs ≤ 0.25; see Table S6) across the three trimesters. As a result, geometric mean values for the three trimesters were calculated to better reflect the average exposure level during pregnancy. Since the concentrations of OPEs and PAEs were generally skewed, their distributions were described using the geometric mean and the 25th−75th percentiles (P_{25}−P_{75}). Spearman's rank correlation analysis was employed to assess pairwise correlations between OPEs and PAEs. Owing to the skewed distribution, concentrations after CR-adjustment were transformed using the natural logarithm (ln) to approximate normality and were used in subsequent analyses, thereby reducing the influence of extreme values. Missing covariate data were imputed using the random forest method. Differences between groups were analyzed using one-way analysis of variance (ANOVA) for three groups, independent t tests for two groups, and chi-square tests for categorical variables.

We used group-based multitrajectory modeling (GBMTM) to analyze the bivariate trajectories of ASD and ADHD symptom scale scores in preschool children at ages 3, 5, and 6. GBMTM enabled the use of a detailed longitudinal data method that employs maximum-likelihood estimation to impute missing data and group individuals with similar patterns of outcomes measured over time.[34] To determine the optimal number of stable trajectories for the multitrajectory model and verify the statistical hypotheses, we initially estimated each dimension with TRAJ in Stata. Model selection should take into account field knowledge to ensure the practical usefulness of the findings.[37] It should also consider the following: (1) the Bayesian information criterion (BIC), where a value closer to zero indicates a better fit; (2) higher entropy, which indicates more apparent separation between trajectory groups; (3) the average posterior probability (AvePP), which should exceed 0.7; and (4) each trajectory group containing more than 5% of participants.[38]

Primary analysis employed a four-step approach. First, restricted cubic splines (RCSs) were used to examine whether a nonlinear relationship existed between prenatal OPEs and PAEs with children's ASD and ADHD symptoms cotrajectory. Second, we conducted stratified multivariate logistic regression analyses by trimester and sex. Third, we used generalized estimating equations (GEEs) to thoroughly assess longitudinal relationships among OPEs/PAEs between ASD/ADHD symptoms. Fourth, we used Bayesian kernel machine regression (BKMR) and quantile-based g-computation (QGC) models to evaluate the association between combined prenatal OPE and PAE exposures and the cotrajectory. To minimize the false-positive rate (Type I error) across multiple comparisons, the false discovery rate (FDR) was used to adjust p-values via the Benjamini−Hochberg method, yielding q-values.

In the metabolomics analysis, the metabolite data were log2-transformed to achieve normality. Owing to the observed sex disparity, we performed MWAS, combined with the MITM method, separately for boys, girls, and all children. First, a multivariate linear regression model was used to identify metabolites associated with exposure, and a multivariate logistic regression model was used to analyze metabolites related to the cotrajectory, adjusting for relevant covariates. Although an FDR threshold of <0.05 was initially applied, fewer associations remained after correction because of the high dimensionality and limited statistical power of the data. To detect potential subtle associations, nominal p-values (<0.05) were used for exploratory metabolite identification.[26] The MITM analysis involved two steps: first, enriched pathways related to exposure and cotrajectory were identified, and second, shared metabolites involved in both associations were identified. Next, the "mediation" package in R was used to further analyze the mediating role of these shared metabolites in the relationship between exposure and cotrajectory. Metabolites showing significant mediation effects were subjected to additional enrichment analysis.

Data analysis was conducted using SPSS 23.0 (SPSS Inc., Chicago, IL), Stata 14.0 (StataCorp, TX, USA), and R (4.1.2). Graphs were created with GraphPad Prism 9.0 (GraphPad Inc., California, USA), and pathway enrichment analysis was performed using the "MetaboAnalyst 6.0" online platform. The significance level was set at 0.05.

2.7. Sensitivity Analysis

To evaluate the robustness of the main results, several sensitivity analyses were conducted. (1) Since the detection rates of BBOEP, BEHP, and TCEPD are below 85%, considering that the result may

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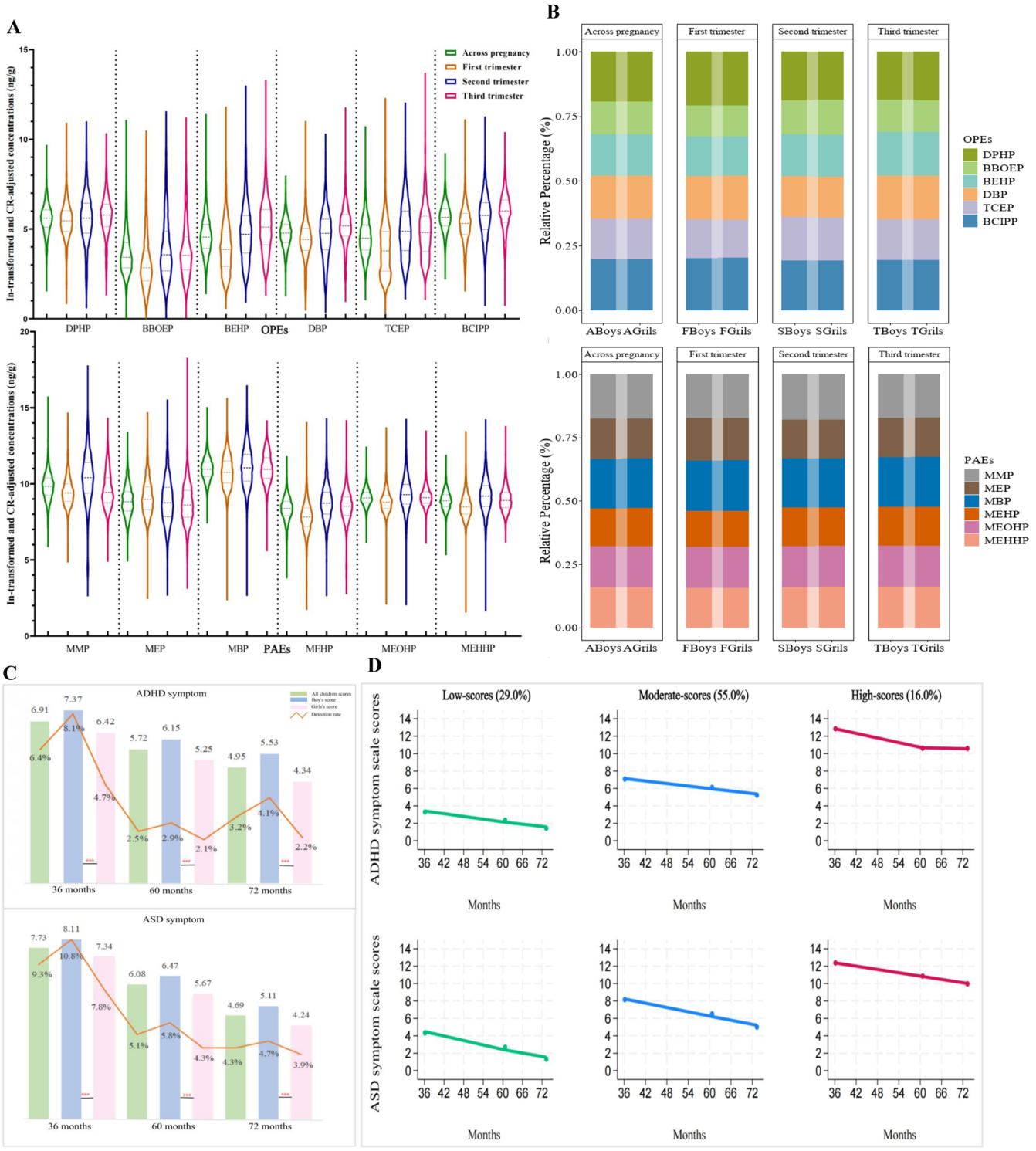


Figure 2. Description of the basic situation of exposures and outcomes. (A) Violin plots showing maternal ln-transformed and CR-adjusted urinary OPE (top in the figure) and PAE (bottom in the figure) metabolite levels across the entire pregnancy and each trimester. Green indicates across pregnancy, orange indicates the first trimester, blue indicates the second trimester, and rose red indicates the third trimester. (B) Composition profiles of maternal ln-transformed and CR-adjusted urinary OPE (top in the figure) and PAE (bottom in the figure) metabolites during pregnancy, categorized by child sex, for different trimesters. (C) Distribution of ADHD (top in the figure) and ASD (bottom in the figure) symptoms in children of different ages and sexes, *p-value <0.05, **p-value <0.01, ***p-value <0.001. The green bars represent all children, blue for boys, pink for girls, and the orange line represents the detection rate. (D) Trajectories of preschoolers' ADHD and ASD symptom scale scores derived from group-based multitrajectory models. The groups are named based on their initial scores and subsequent trends. The green line indicates the low-scores trajectory, the blue line indicates the moderate-scores trajectory, and the red line indicates the high-scores trajectory.

be affected by the substitution of LOD/$\sqrt{2}$, we first substitute LOD/2 to make the calculation.[36] Then we regard the data for these three substances as < LOD and use the missing-indicator method for analysis.[39] (2) Exclude premature infants (3) Adjustments were made

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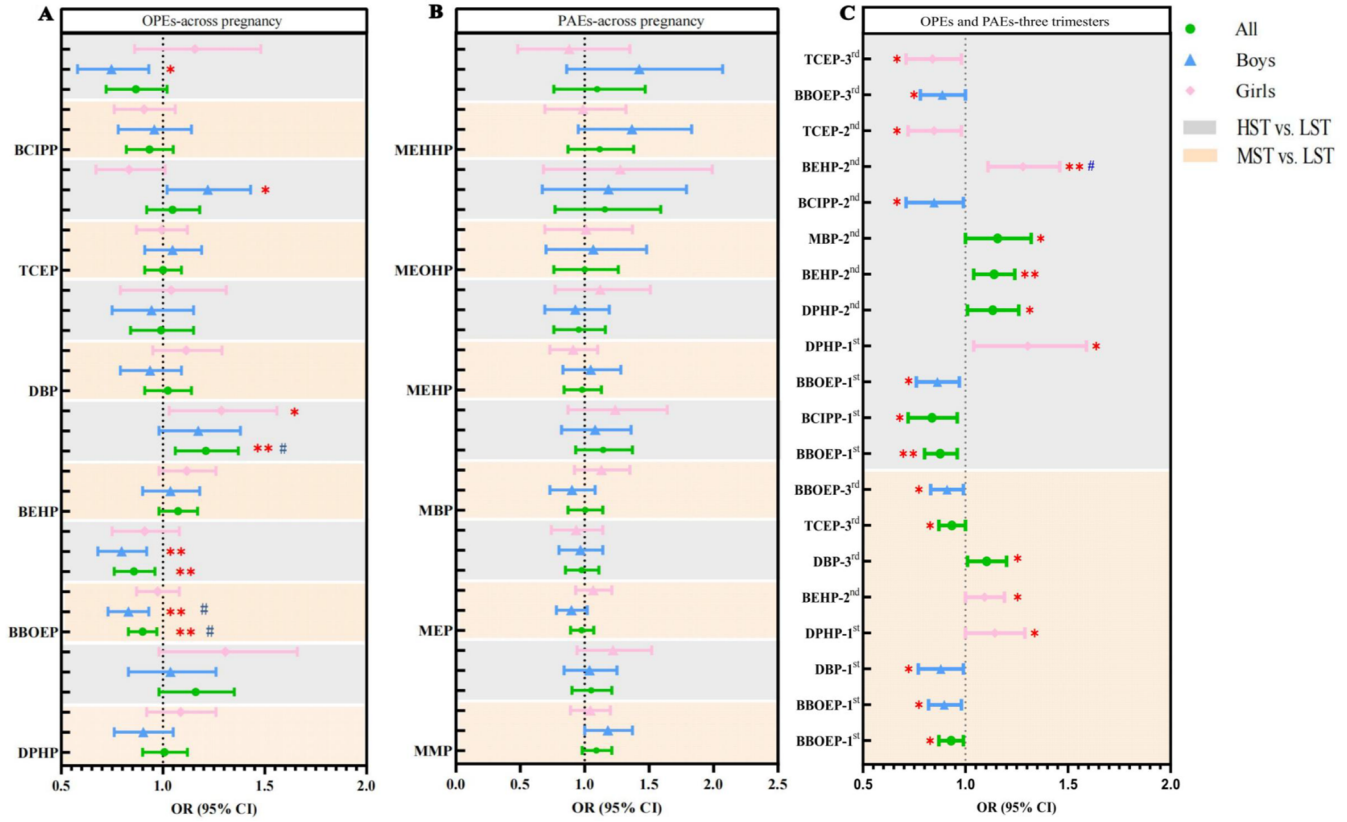


Figure 3. Associations between individual OPEs and PAEs across pregnancy and different trimesters with the cotrajectory of ADHD and ASD symptoms scale scores were analyzed using logistic regression. (A) Associations of individual average OPEs across pregnancy with the cotrajectory of ADHD and ASD symptoms scale scores. (B) Associations of individual average PAEs across pregnancy with the cotrajectory of ADHD and ASD symptoms scale scores. (C) Associations of individual OPEs and PAEs in different trimesters with the cotrajectory of ADHD and ASD symptoms scale scores (p-value <0.05). Green indicates all children, blue indicates boys, pink indicates girls, and the gray background shows HST versus LST, while the yellow background shows MST versus LST. Models were adjusted for maternal age, prepregnancy BMI, gestational weight gain, per capita monthly family income, maternal educational level, delivery mode, parity, maternal IQ, infant sex, and breastfeeding mode. Across pregnancy is defined as the geometric mean concentration for the three trimesters. Significant associations are marked with asterisks (*p-value < 0.05, **p-value < 0.01, #q-value < 0.05; q-values were corrected by the Benjamini-Hochberg method). LST, Low-scores trajectory; MST, Moderate-scores trajectory; HST, High-scores trajectory; 1st, first trimester; 2nd, second trimester; 3rd, third trimester.

to account for maternal smoking and drinking before or during pregnancy, based on the original confounding factors, and then for folic acid supplementation. (4) We further performed trajectory modeling for the symptom scale scores of ADHD and ASD to compare the similarities and differences with the results of the cotrajectory analysis.

3. RESULTS

3.1. Participant Characteristics

Among the 3040 mother−child dyads included in our study, the mean maternal age was 26.66 years, and most mothers (68.9%) had a prepregnancy BMI within the normal range. The majority of mothers (80.5%) had completed high school or higher, and most were primiparous (88.7%). The study group included 1556 boys and 1484 girls. Significant differences (p < 0.05) in prepregnancy BMI, educational level, parity, and IQ scores were detected between the included (n = 3040) and excluded (n = 434) mother−child dyads, as shown in Table S5.

3.2. Maternal OPE Concentration and Variability

As shown in Table S6, the detection rates for OPE metabolites ranged from 73.7% to 95.5%, with DPHP achieving the highest rate. The PAE detection rates exceeded 98% for all the compounds, with MEHP showing the lowest rate. For both uncorrected and CR-adjusted concentrations, the ICC for all OPE and PAE metabolites was ≤0.25, indicating poor reproducibility and high variability over time. The ln-transformed and CR-adjusted concentration distributions of OPEs and PAEs are shown in Figure 2A, both of which are normal. There were no significant differences in OPEs and PAEs between boys and girls (p > 0.05) (Figure 2B and Table S7). OPEs were significantly weakly correlated throughout pregnancy and across trimesters (0.074 ≤ rs ≤ 0.393); PAEs were significantly correlated (0.106 to 0.848); and most correlations between OPEs and PAEs were statistically significant (p < 0.05, 0.440 ≤ rs ≤ 0.481), as presented in Table S8 and Figure S3.

3.3. Basic Conditions and Cotrajectory of ADHD and ASD Symptoms

As shown in Figure 2C and Table S9, the detection rates of ADHD symptoms at 3, 5, and 6 years of age were 6.4%, 2.5%, and 3.2%, respectively, whereas the detection rate of ASD symptoms gradually decreased with age, with values of 9.3%, 5.1%, and 4.3%, respectively. Boys consistently exhibited higher rates of ADHD and ASD symptoms than girls (Figure 2C). By combining model analysis with practical significance,

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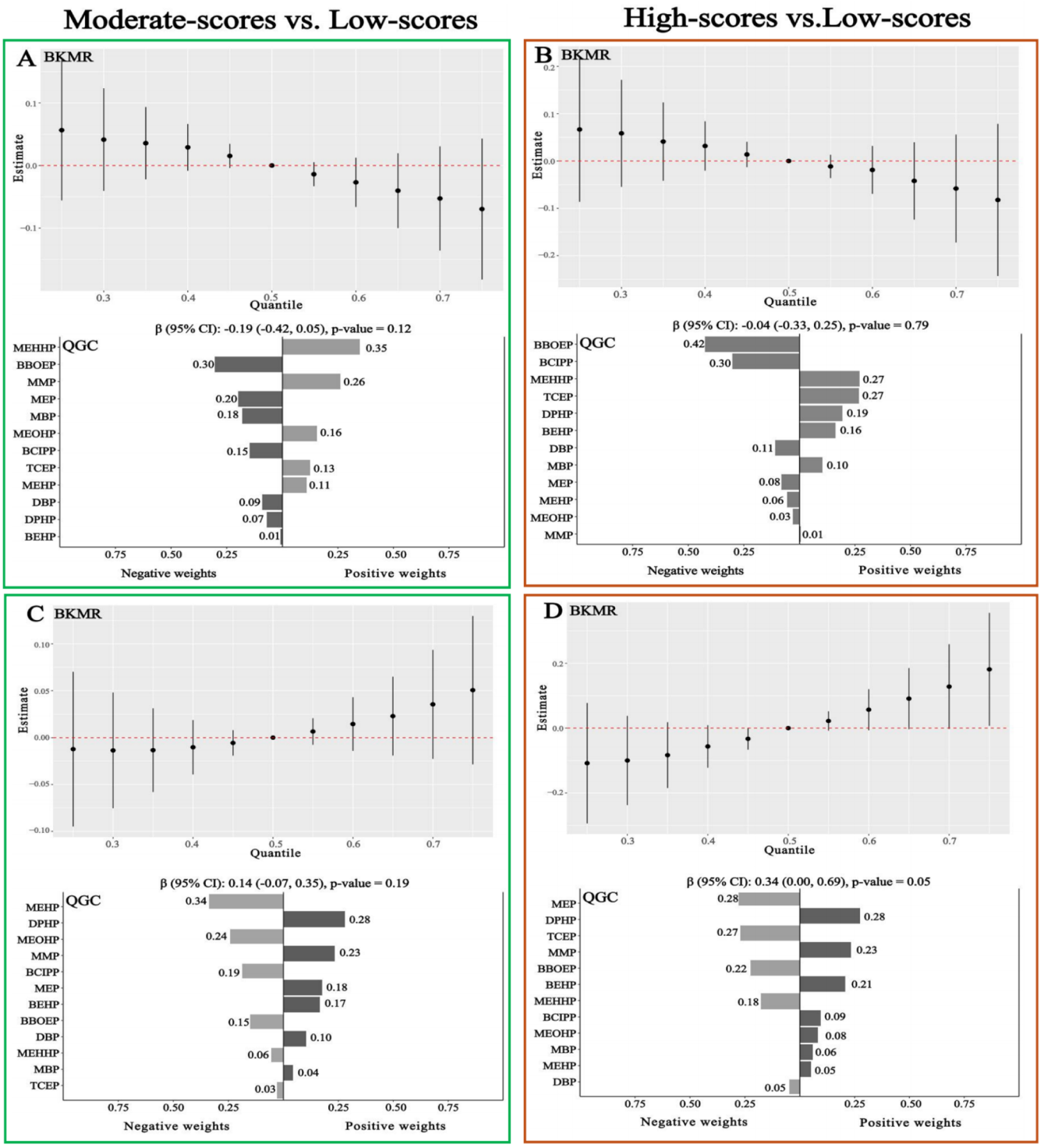


Figure 4. Associations of OPEs and PAEs mixture across pregnancy with the cotrajectory of ADHD and ASD symptoms scale scores estimated from Bayesian kernel machine regression and quantile g calculation (stratified by sex). (A) The association between mixed exposure and moderate-scores trajectory among boys. (B) The association between mixed exposure and high-scores trajectory among boys. (C) The association between mixed exposure and moderate-scores trajectory among girls. (D) The association between mixed exposure and high-scores trajectory among girls. Model was adjusted for maternal age, prepregnancy BMI, gestational weight gain, per capita monthly family income, maternal educational level, delivery mode, parity, maternal IQ, and breastfeeding mode; across pregnancy is defined as the geometric mean concentration for the three trimesters.

we found that although the BIC and AIC values for the 4-classification model were closer to zero, the model's entropy was lower, and the practical significance of the 4-classification groups was limited. Therefore, we determined that the typical trajectories of ADHD (order 1 1 2) and ASD (order 1 1 1) symptom scale scores could be divided into 3 groups (Table S10). The three groups are presented in Figure 2D: "Low-scores" trajectory (LST) (29.0%), "Moderate-scores" trajectory

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(MST) (55.0%), and "High-scores" trajectory (HST) (16.0%) (HST indicated that children have high scores on both the ADHD and ASD scales concurrently at each time point; the same applies to the other groups; for details see Table S11). Table S12 compares the demographic characteristics among the different trajectory groups. The results revealed differences among the groups in terms of maternal age, education level, per capita monthly family income, maternal smoking and alcohol consumption before or during pregnancy, maternal IQ, child sex, and feeding mode.

3.4. Associations between OPE and PAE Exposure with Cotrajectory

3.4.1. Individual OPE and PAE Exposure. *3.4.1.1. Nonlinear Associations Between Average OPEs and PAEs with Cotrajectory.* Using RCS analysis (Figures S4−S5), we further explored the dose−response relationship between the two substances throughout pregnancy and their cotrajectory after adjusting for covariates. For OPEs, we found a nonlinear relationship between BBOEP ($p_{nonlinear}$ = 0.047) and DBP ($p_{nonlinear}$ = 0.007) with the HST. The results for PAEs revealed a nonlinear relationship between MEHP and the HST ($p_{nonlinear}$ = 0.027); however, no statistically significant nonlinear relationship was observed for the other metabolites ($p_{nonlinear}$ > 0.05).

3.4.1.2. Associations between Average OPEs and PAEs with Cotrajectory Are Shown. The relationships between the average concentrations of OPE and PAE metabolites across pregnancy and cotrajectories are shown in Figure 3A−B and Table S13. Among OPEs, higher DPHP concentrations were associated with an increased risk of being in the HST (OR = 1.20, 95% CI: 1.03, 1.40). However, this association was no longer significant after adjusting for covariates. BBOEP was associated with decreased odds of the MST and HST (q < 0.05), both before and after covariate adjustment. BEHP was associated with an increased risk of HST [OR (95% CI): before, 1.16 (1.02, 1.31); after, 1.20 (1.06, 1.37), q < 0.05], with significance maintained after adjustment for multiple comparisons (Figure 3A). No statistically significant associations were observed for PAEs (Figure 3B).

3.4.1.3. Sex Stratification Analysis of Average OPEs and PAEs with Cotrajectories. The logistic regression model, after adjusting for covariates, revealed that in boys, BBOEP exposure throughout pregnancy was negatively associated with MST (OR = 0.83, 95% CI: 0.73, 0.93) and HST (OR = 0.79, 95% CI: 0.68, 0.92; q < 0.05); BCIPP was also negatively associated with HST among boys ($p_{sex\text{-}int}$ < 0.05). However, among girls, only BEHP was associated with an increased risk of HST (OR = 1.27, 95% CI: 1.03, 1.56). For detailed information, please refer to Figure 3A and Table S14.

3.4.1.4. Trimester Stratification Analysis of OPEs and PAEs with Cotrajectories. After adjusting for covariates, compared with the LST, BBOEP in the first trimester was negatively associated with the MST (OR = 0.93, 95% CI: 0.87, 0.99) and HST (OR = 0.87, 95% CI: 0.80, 0.96), and BCIPP was negatively associated with the HST (OR = 0.83, 95% CI: 0.72, 0.96); second-trimester DPHP (OR = 1.13, 95% CI: 1.01, 1.26), BEHP (OR = 1.14, 95% CI: 1.04, 1.24), and MBP (OR = 1.15, 95% CI: 1.00, 1.32) were positively linked to the HST, and third-trimester DBP was positively related to the MST (p < 0.05), while TCEP was negatively associated (p < 0.05) (Figure 3C and Table S15).

We also observed sex- and trimester-specific disparities (Figure 3C and Table S16). After adjusting for covariates, in boys, BBOEP in the first and third trimesters was negatively associated with MST and HST compared with LST, and DBP in the first trimester was negatively associated with MST. In the second trimester, BCIPP was negatively linked to HST (p < 0.05). In girls, first-trimester DPHP was positively associated with MST and HST ($p_{sex\text{-}int}$ < 0.05); in the second trimester, BEHP was positively associated with MST and HST (q < 0.05), whereas TCEP was negatively associated with HST in the second and third trimesters.

3.4.2. Exposure to OPEs and PAEs Mixtures. After adjusting for covariates, the BKMR model indicated that mixed exposure to OPE and PAE metabolites across pregnancy and during the three trimesters was not significantly associated with cotrajectory (Figure S6). We further analyzed exposure by sex across pregnancy and found no mixture effect. However, an interesting pattern emerged: a negative association was observed among boys (Figure 4A−B), whereas the opposite was observed among girls (Figure 4C−D). Additionally, among girls, when the combined concentration exceeded the P_{75}, the risk of HST might increase, and this association was statistically significant (Figure 4D). The QGC model results aligned with those of the BKMR model (Figure 4A−D).

3.5. Associations of OPEs and PAEs with ADHD and ASD Symptoms among Preschoolers

The results from the GEE model closely matched those from the trajectory analysis (Figure S7 and Tables S17−S20). With respect to ADHD symptoms scale scores, $BBOEP_{avg}$ (β = −0.14, 95% CI: −0.27, −0.01) and $BCIPP_{first}$ (β = −0.18, 95% CI: −0.35, −0.02) were negatively associated, whereas $BEHP_{avg+second}$ and MBP_{second} were positively associated (p < 0.05); with ADHD symptoms, DBP_{third} was negatively associated (OR = 0.84, 95% CI: 0.73, 0.96); TCEP, BCIPP, MEP, MEHO, MEOHP, and MEHHP show sex-specificity in the aforementioned correlation (all $p_{sex\text{-}int}$ < 0.05). With respect to ASD symptoms scale scores, $DBP_{avg+second}$ were negatively associated, whereas $BEHP_{avg+second}$, $MEOHP_{avg+third}$, and MMP_{third} were positively associated (p < 0.05); with ASD symptoms, $MEP_{avg+second}$, DBP_{second}, and MBP_{third} were negatively associated, whereas $BEHP_{avg+second+third}$ and MMP_{avg} were positively associated (p < 0.05); DPHP, BBOEP, DBP, TCEP, BEHP, and BCIPP show sex-specificity in the aforementioned correlation (all $p_{sex\text{-}int}$ < 0.05).

3.6. Metabolites Associated with OPE and PAE Exposure with Cotrajectories

After quality control, 4632 cord blood metabolic profiles remained for analysis. The MWAS results revealed that among all the children and boys, the number of cord blood metabolites associated with MEHP was highest, whereas MBP was most abundant in girls (Figure S8). After FDR adjustment (q < 0.2), only a few metabolic features remained significant (Table S21). Therefore, we used a nominal p-value threshold of <0.05 to identify potentially relevant metabolic features for further exploratory analysis. Logistic regression identified 456 and 594 metabolites significantly associated with MST and HST, respectively, across all children; 478 and 576 in boys; and 328 and 546 in girls (Table S21).

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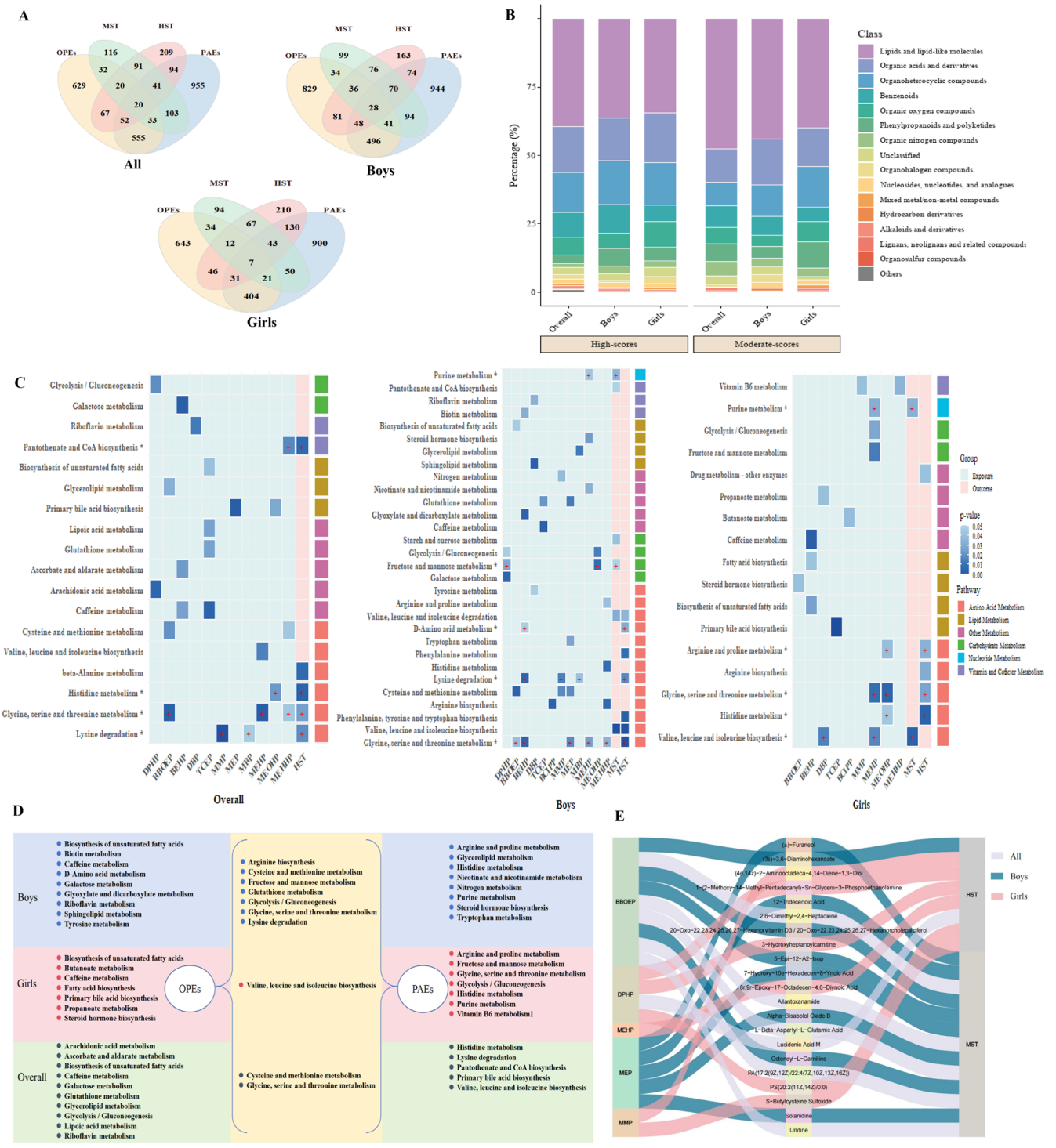


Figure 5. Metabolomics analysis of the association of OPEs and PAEs across pregnancy with the cotrajectory of ADHD and ASD symptoms scale scores. (A) The Venn diagram illustrates the overlap of significant features between OPEs and PAEs exposure across pregnancy (linear regression models) and the cotrajectory of ADHD and ASD symptoms scale scores (logistic regression models). (B) The proportion of metabolites shows significant cross-interaction between exposure and outcome. (C) Heat map of p-values for enrichment pathways (p-values <0.05). Each cell in the heatmap shows the p-value for pathways linked to OPEs, PAEs, or cotrajectory. Overlapping pathways are marked with a red plus sign (+). (D) Enrichment pathways associated with at least one chemical, including pathways unique to OPEs and PAEs exposure, as well as those shared between the two chemicals. (E) The Sankey plot illustrates the mediating role of cord blood metabolites in the association between chemicals and cotrajectory (with p-values for the total effect and mediating effect <0.05). The first node represents the chemical exposure, connected via relevant metabolites to the second node, which ends at the cotrajectory in the last node. Different colors on the left represent various chemicals, and the colors on the right indicate different score trajectories (MST or HST). The purple connection represents all children, blue represents boys, and pink represents girls. MST, Moderate-scores trajectory; HST, High-scores trajectory.

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3.7. Overlapping Enriched Pathways Associated with Both OPE and PAE Exposure and Cotrajectories

The results of the pathway analysis are shown in Figure 5C−D and Table S22. Across all children, metabolites associated with OPEs were primarily enriched in 12 metabolic pathways. In contrast, PAEs were enriched in 7 pathways, with two overlapping pathways (cysteine and methionine metabolism and glycine, serine, and threonine metabolism). For HST, 5 pathways were enriched, four of which were amino acid metabolic pathways. Four pathways were enriched for both exposure and outcome, which were primarily represented by amino acid metabolic pathways (Figure 5C−D and Table S22).

In boys, metabolites related to OPEs were enriched mainly in 16 metabolic pathways, whereas those related to PAEs were enriched in 15 pathways, with 7 pathways overlapping. With respect to HST, 7 pathways were enriched, all of which involved amino acid metabolism pathways; regarding MST, 6 pathways were enriched. Five pathways were enriched for both exposure and outcome, which were represented primarily by amino acid metabolic pathways (Figure 5C−D and Table S22).

In girls, metabolites related to OPEs were enriched in 8 metabolic pathways, whereas those related to PAEs were enriched in 8 pathways, with only one pathway overlapping (valine, leucine, and isoleucine biosynthesis). For HST, 5 pathways were enriched, and for MST, 2 pathways were enriched. Five pathways were enriched for both exposure and outcome, which were represented primarily by amino acid metabolic pathways (Figure 5C−D and Table S22).

In all the children, BBOEP, MMP, MBP, MEHP, MEOHP, and MEHHP all had overlapping pathways with HST. In boys, BBOEP, BEHP, MMP, MEP, MBP, MEHP, and MEHHP all had overlapping pathways with HST, whereas DPHP, MEHP, and MEOHP had overlapping pathways with MST. In girls, only MEHP and MEOHP had overlapping pathways with HST, whereas DBP and MEHP had overlapping pathways with MST (Figures 5C and Table S22). From the above finding, we can see an apparent sex disparity, and OPEs and PAEs may cause neurodevelopmental disorders in offspring via the same metabolic pathways.

3.8. Overlapping Metabolites Associated with Both OPE and PAE Exposure and Cotrajectories

The results of the exploratory MITM analysis revealed that the number of cross-metabolites related to OPEs and MST in all children, boys, and girls were 105, 139, and 74, respectively; the numbers of OPEs and HST were 159, 193, and 96, respectively; the numbers of PAEs and MST were 197, 233, and 121, respectively; and the numbers of PAEs and HST were 207, 220, and 211, respectively (Figure 5A and Table S23). After these metabolites were classified, the top three rankings were the same across all the children and, for sex stratification, consisted of lipids and lipid-like molecules, organic acids and derivatives, and organoheterocyclic compounds (Figure 5B). A total of 129 metabolites showed significant mediation in the analysis of all children (105 in the mediation analysis of boys, and 93 in the mediation analysis of girls; Tables S24−S26). We further required that both the total and the mediation effects be significant; only 6 metabolites were significant in all children, 9 in boys, and 6 in girls (Figure 5E). Pathway enrichment analysis of these biomarkers revealed that pyrimidine metabolism mediated exposure to OPEs and the cotrajectory of all children, whereas biotin metabolism and lysine degradation mediated exposure to OPEs and the cotrajectory of boys (Table S27). Cysteine and methionine metabolism mediated exposure to OPEs, whereas purine metabolism mediated exposure to PAEs and the cotrajectory of girls (Table S27).

3.9. Sensitivity Analyses

We replaced data < LOD with LOD/2, and using the missing-indicator method for OPEs with a detection rate below 85%, the results remained unchanged (Table S28). Further excluded preterm infants, and even after controlling for smoking, alcohol consumption during pregnancy, and folic acid supplementation, the results remained robust (Table S29). Both ADHD and ASD symptom scale scores ultimately follow three trajectories (Table S10 and Figure S9). Some differences were detected between the results for the separate trajectory and the cotrajectories: the effects of prenatal OPE and PAE exposure on the common trajectory were more pronounced (Table S30).

4. DISCUSSION

On the basis of 3040 mother-child dyads in the MABC, increased BEHP exposure across pregnancy was associated with a higher risk of HST among preschoolers. BBOEP exhibited significant nonlinear and U-shaped associated HST. Sex-specific disparities were identified. The first-to-second trimester may emerge as a sensitive window. No mixed effects were observed. Prenatal exposure to OPEs and PAEs may be further linked to offspring neurodevelopment through the disruption of cord blood metabolomics.

Overall, the concentrations of OPEs detected in pregnant women in our study were comparable to those reported in other Chinese provinces but lower than those reported in Canada and the United States. For a detailed summary, please refer to the previous literature.[32] The PAE concentrations in this study also align with those in the most recent Chinese research but differ from those reported in other countries. In summary, the detection rates and patterns for both OPEs and PAEs vary across nations, indicating regional-specific exposure scenarios in which disparities may influence national policies, dietary habits, lifestyles, detection methods, and sampling times. For detailed information, please refer to the cited literature.[33]

The detection rates of ADHD symptoms at ages 3, 5, and 6 were 6.4%, 2.5%, and 3.2%, respectively. These rates are lower than the global prevalence of 7.2% reported in a meta-analysis.[40] North America had the highest prevalence (6.2%), followed by Europe (4.7%), whereas the Middle East had the lowest prevalence (2.3%).[41] In Italy, the prevalence among 5 to 17-year-olds ranged from 1.1% to 3.1%, with boys' prevalence higher than girls', consistent with our findings.[42] ASD symptom detection rates (9.3%, 5.1%, 4.3%) are higher than the global meta-analytic estimate of 1.19%.[43] Recent research has also indicated that the highest estimated prevalence occurs in high-income super-regions, with an estimated value of approximately 1090.2 per 100000 people.[3] Our higher prevalence reflects screening-based assessment rather than clinical diagnosis—a trade-off enhancing efficiency in large-scale epidemiology.[32,34] Age also drives these variations: ASD emerges with early signs (e.g., language and social delays) at ages 2−3, yielding higher rates in children under 5. Age also drives these variations,[44] primarily because ASD typically

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manifests with early signs—such as language delays and social withdrawal—that usually become apparent between ages 2 and 3, leading to higher diagnostic rates among children under 5,[45] in contrast, inattention/hyperactivity is often missed in children under 5, where behavioral expectations are lower.[11] However, entering school, increased academic and behavioral demands make these symptoms more apparent, leading to higher diagnosis rates.[46] This phenomenon explains our higher detection rate of ASD than ADHD. Moreover, we reported that ASD detection rates declined with age, aligning with prior reports.[45] Because reliance on parental reports may inevitably be influenced by parents' educational and cultural backgrounds, necessitating cautious interpretation. Our reliance on questionnaire-based scales limits the ability to draw definitive conclusions, underscoring the need for future validation using clinical diagnostics. Due to the questionnaire and the age group, the rate of detected ASD symptoms was high, which may further affect the generalizability of the results.

One study reported that maternal exposure to DPHP in the third trimester was associated with an increased risk of ADHD in children.[22] Moreover, BBOEP showed no association, with the effect being more pronounced in girls.[22] Another study indicated that maternal exposure to di-*n*-butyl phosphate and bis (1,3-dichloro-2-propyl) phosphate in the first trimester was positively correlated with ADHD among preschool children. As BBOEP exposure increased, the risk decreased among girls but increased among boys.[23] A study revealed that maternal di (2-ethylhexyl) phthalate exposure in the third trimester increased the odds ratio for ASD among 2-year-olds.[47] Prenatal MEHHP exposure was also associated with increased ADHD symptoms in children aged 2.5 and 11 years (OR = 2.98) in both sexes.[48] Kim et al. further demonstrated a positive association between MEHHP and ASD traits in 4-year-old children, with stronger correlations observed in boys.[49] Prenatal exposure to mixed PAEs was also positively correlated with behavioral disorders such as ADHD.[50] Concurrently, our prior research indicated that prenatal exposure to PAEs increases offspring risk for ASD symptoms, with notable sex disparities.[33] We observed a negative correlation between BBOEP and the cotrajectories. This phenomenon may be due to the nonlinear effect of BBOEP suggested by the Maternal And Developmental Risks from Environmental and Social Stressors study.[51] Our RCS analysis revealed that, in the high-score trajectory, significant nonlinear and U-shaped patterns emerged during maternal BBOEP exposure. This finding explains why some compounds are positively correlated while others are negatively correlated. Future explanations of how exposure influences health outcomes should consider differences in dose and chemical potency. Few studies have combined OPE and PAE exposure. Our prior work confirmed that combined prenatal exposure to OPEs and PAEs impairs offspring cognition.[13] Although no overall population combined effect was observed, girls exposed above the 75th percentile showed an increased risk of HST.

The underlying mechanisms of OPE/PAE that affect children's neurodevelopment remain unclear. Animal studies link environmental OPE/PAE exposures to altered early behavioral development via multiple pathways (inflammation, neurotransmitter systems, oxidative stress, endocrine disruption).[51] A study found that OPEs reduced glutamate and dopamine, triggering inflammation and disrupting myelin formation.[52] The placenta mediates OPE neurotoxicity, with greater accumulation in male placentas driving reduced prefrontal serotonin, inflammation, and endocrine disruption.[51] OPE/PAE metabolites may disrupt fetal neurodevelopment by interfering with maternal thyroid hormone levels.[13,33,34] PAE exhibit complex sex hormone interactions (estrogenic/androgenic effects), likely driving the observed sex-specific responses, and vary with exposure timing. For example, early exposure to endocrine-disrupting compounds is strongly associated with greater male behavioral deficits, while females may be more susceptible to stress encountered in later life.[53] Sex-specific trajectories underscore the need for sex-stratified analyses to avoid misclassifying risk and to inform targeted public health interventions. Our results showed that different chemicals present different effects and patterns. These inconsistencies may be attributed to various factors, such as misclassification of exposure, critical exposure windows, sex-specific effects, and the influence of other mixtures.[53]

Our findings demonstrate that prenatal OPE/PAE exposures disrupt cord blood lipid metabolism and accumulation, corroborating prior research on their established effects on offspring's physical development.[32,54] Pathway analysis linked both OPEs and PAEs predominantly to amino acid metabolism, notably cysteine and methionine metabolism/glycine, serine, and threonine metabolism, with enhanced convergence observed specifically in male offspring. Exposure−outcome pathway overlaps revealed histidine metabolism/glycine, serine, and threonine metabolism/lysine degradation/pantothenate and CoA biosynthesis.

Mediation analysis revealed that pyrimidine metabolism mediated the effects of OPEs across all children. Sex-specific pathways included biotin metabolism and lysine degradation (boys, OPEs) and cysteine/methionine and purine metabolism (girls, OPEs/PAEs). These mediation pathways are biologically coherent. Purine and pyrimidine metabolism are fundamental to brain development, where nucleotide imbalances disrupt phospholipid synthesis and membrane integrity, linking OPEs/PAEs to neurodevelopmental risk.[54−57] Biotin is an essential cofactor for neurodevelopment, supporting neuronal energy metabolism and gene regulation via histone biotinylation;[58−61] its role aligns with our prior finding that prenatal vitamin supplementation mitigates environmental neurotoxicity.[13,33,34] Lysine degradation provides acetyl-CoA for myelin and cholesterol synthesis, while cysteine/methionine metabolism maintains glutathione-dependent antioxidant defenses critical for protecting the lipid-rich developing brain.[62−65] Together, these pathways suggest that prenatal OPEs and PAEs exert neurotoxic effects primarily by disrupting metabolic homeostasis rather than through direct neurotoxicity, consistent with prior research findings.[15,25−27]

Our study has several strengths, including a prospective birth cohort design and a comprehensive collection of gestational data. An integrated metabolomics framework was employed to identify overlapping pathways and metabolites involved in the association between maternal urinary OPEs and PAEs and the cotrajectory of preschoolers' ADHD-ASD symptoms. This approach enabled a holistic evaluation of all metabolites detected within pathways to assess potential mediating pathways. Furthermore, our study included multiple outcome assessments and conducted cotrajectory analyses, offering novel insights for future research. The relatively large sample size of our study also enhances the robustness of the outcome findings.

Our study has several limitations. First, residual confounding factors—such as exposure to OPEs and PAEs during

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childhood, physical activity, and sleep patterns—may remain in observational studies and could affect the observed associations. Second, two scales were used to evaluate ADHD and ASD symptoms in children. However, while these scales are suitable for large-scale epidemiological surveys, their sensitivity and specificity are limited. Additionally, among the 3040 mother−child dyads included in our study, 94.9%, 60.4%, and 70.3% of the children underwent follow-up for ADHD symptoms at ages of 3, 5, and 6 years, respectively, while 95.1%, 60.5%, and 69.9% underwent follow-up for ASD symptoms at these ages. More than 30% of the children lacked available scores, which could introduce bias when interpreting the model results stratified by age. Therefore, comparisons across age-stratified models should be approached with caution. Third, in this observational exploratory study, the use of an uncorrected significance threshold in metabolomics analysis may increase the Type I error rate, but this was a necessary compromise to balance the likelihood of missed discoveries and the risk of false discoveries.[26] These results should still be considered preliminary and exploratory, and they should be interpreted with caution. Fourth, the metabolite detection method employed in this study utilized a non-targeted metabolomics strategy. Fifth, we employed the ACQUITY UPL CHSST3 column for metabolite separation, which exhibited poor retention of both nonpolar and polar compounds, potentially resulting in fewer metabolites being detected.[66,67] Finally, owing to variations in geographic location, lifestyle habits, and regional levels of control over environmental pollutants, exposure levels to OPEs/PAEs differ among populations.

Overall, the results of this study support the current mechanistic understanding of how prenatal exposure to OPEs and PAEs may be linked to neonatal metabolomics and how these biological disruptions could influence neurobehavioral outcomes. Further research is needed to assess the generalizability of these findings, examine differences across populations, incorporate additional coexposures, and explore the potential of these metabolic responses for detection, treatment, and prevention strategies in public health and clinical settings.

■ ASSOCIATED CONTENT

Data Availability Statement

Data from all participants in this study were anonymized and accessible only to authorized members of the study team.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.6c01402.

Sample preparation, processes for measuring OPEs and PAEs exposure in urine, QA/QC, metabolomics detection in cord blood, data preprocessing, and additional tables and figures (PDF)

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Notes

The authors declare no competing financial interest.

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